SEMESTER PROJECT REPORT

BUILT-IN SELF-TEST (BIST)
4-BIT BINARY ADDER

AUTONOMOUS TEST-PATTERN GENERATION RESPONSE COMPACTION SIGNATURE ANALYSIS

Logic Circuit Design Lab

Institution: Govt. Model Engineering College, Thrikkakara

Course: Logic Circuit Design Lab

Circuit Under Test: 74LS83A 4-bit binary full adder

FPGA target selected in Vivado: Xilinx Spartan-7 XC7S50CSGA324-1

Project team

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Prepared using the supplied project report, presentation, IC pinout reference and the Vivado 2024.2 simulation / implementation results shown during the project workflow.

Document guide

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Implementation-status note

The supplied references describe a discrete 74-series hardware BIST demonstrator. In addition, the project was brought into Vivado 2024.2 using Verilog, XDC and Tcl files. The observed behavioral simulation completed all four testbench cases successfully; synthesis and implementation were also reported complete. The available screenshots do not establish that a bitstream was generated or that the physical FPGA board was programmed. This report distinguishes the discrete-logic architecture from the FPGA tool-flow results and does not claim completed physical FPGA validation.

1. Abstract

Built-In Self-Test (BIST) is a design-for-testability technique in which a circuit contains or is paired with logic that generates test patterns, observes the circuit response, compresses the response and decides whether the circuit behaves as expected. This project develops a BIST architecture for a 4-bit binary adder, identified in the supplied material as the 74LS83A. A pseudo-random sequence generator supplies operand patterns to the adder. The five adder outputs (four sum bits and carry-out) are combined through an XOR parity network, and the resulting response stream is accumulated in a 4-bit signature register. At the end of the test sequence, a comparator checks the signature against a stored golden value. A normal response is represented by signature 1110; a mismatch produces a FAIL indication. A manual fault switch demonstrates error detection by inverting a sum bit.

The supplied discrete-hardware report describes a 127-cycle test, a 4-bit signature, a golden code of 1110, and a reported 31-of-31 detection result for the evaluated single stuck-at conditions. The Verilog testbench output showed the expected signatures for a fault-free case, an injected-fault case, a deliberately incorrect golden reference and a subsequent fault-free case. Synthesis and implementation completed for the selected Spartan-7 device. The work demonstrates pattern generation, response compaction, signature analysis, comparison and automated test termination.

Keywords: BIST, LFSR, 4-bit adder, XOR compaction, signature register, fault injection, Verilog, Spartan-7, Vivado.

2. Introduction and motivation

Digital systems are built from logic blocks whose correct operation must be verified during development and, in many applications, during manufacturing or maintenance. Conventional external testing relies on Automatic Test Equipment (ATE) to apply input vectors and measure outputs. Such equipment can be expensive, and testing becomes more difficult as circuits grow in complexity or become embedded within larger systems.

BIST addresses part of this problem by incorporating test generation and response evaluation into the system. A BIST controller applies a known sequence of patterns to a circuit under test (CUT), compresses the resulting responses and compares the final compacted value with a reference signature. This reduces the need to observe every output for every test vector. The 4-bit adder is a manageable CUT through which the core BIST concepts can be demonstrated.

3. Aim, objectives and scope

Aim: To design, simulate and demonstrate a BIST architecture for a 4-bit binary adder using pseudo-random test-pattern generation, response compaction, signature analysis and golden-signature comparison.

Objectives

• Generate a repeatable test sequence using an LFSR-based pattern generator.

• Apply generated operand bits to the 4-bit adder under test.

• Combine sum and carry outputs into a response bit using an XOR parity tree.

• Accumulate the response stream in a 4-bit signature register.

• Compare the final signature with golden reference code 1110.

• Provide visible PASS / FAIL status and manual fault injection.

• Verify the Verilog implementation using a testbench and Vivado design flow.

Scope: The report covers the discrete 74-series architecture documented in the supplied references and the subsequent FPGA-oriented Verilog implementation. It reports only simulation and tool-flow milestones visible in the project evidence. Physical programming and board-level FPGA validation remain unconfirmed.

4. BIST fundamentals

A basic BIST system contains a test-pattern generator, a circuit under test, a response compactor/signature analyzer and a test controller with a pass/fail decision. Here, the pattern generator is based on a linear-feedback shift register (LFSR), the CUT is a 4-bit adder, the response compactor is an XOR network, and the signature analyzer is a 4-bit shift register. A comparator evaluates the final signature.

Test-pattern generation: An LFSR is a shift register in which selected stages are combined through XOR or XNOR feedback. The feedback bit is inserted into the register as it shifts, creating a deterministic sequence that can appear pseudo-random.

The same initial state and feedback logic reproduce the same sequence.

Response compaction: An XOR network combines several output bits into one parity bit per test cycle. This reduces the amount of response data that must be stored or compared.

Signature analysis: The compacted response bits are accumulated over the test interval. The resulting multi-bit value is the signature. The expected fault-free signature is the golden signature.

Aliasing: A faulty circuit may produce the same final signature as a fault-free circuit. For an ideal n-bit signature under a simplified random-error assumption, the probability is approximately 1/2■. For n = 4, this is 1/16 = 6.25%.

5. System architecture and operation

The sequence begins when START is activated. The controller enables the clock to the pattern generator and signature register. The LFSR advances through the prescribed test sequence and drives the adder operands. The adder output bits pass through the parity compactor and enter the signature register. At the terminal count, the controller stops the test and enables comparison / result logic.

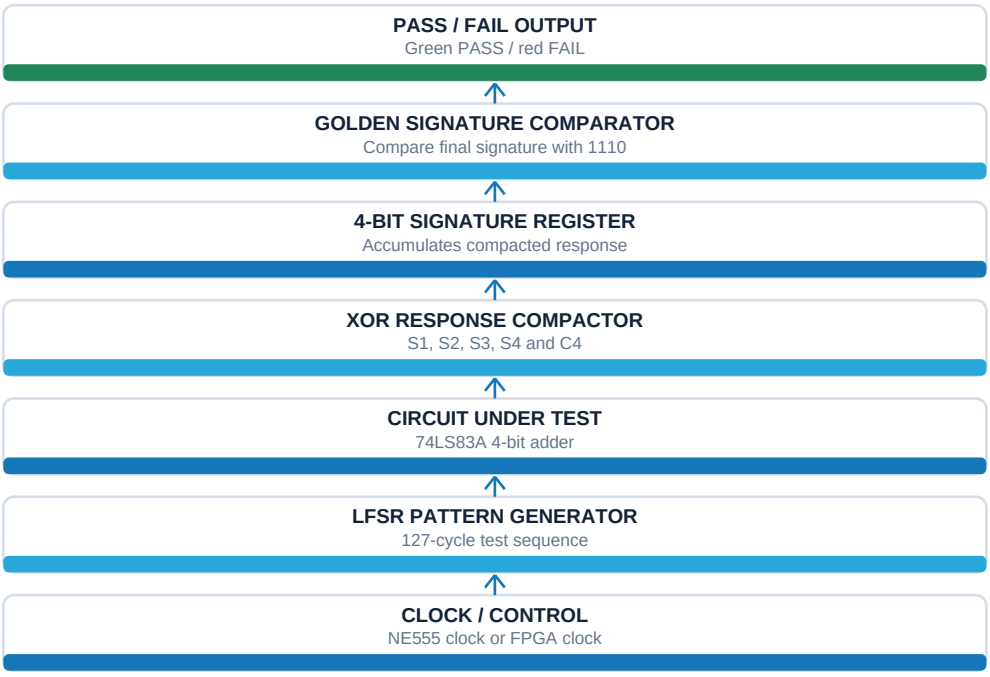


Figure 1. Functional BIST signal flow. This is a conceptual summary; clock and control implementation differs between the discrete IC and FPGA versions.

5.1 Functional sequence

• Initialization: START initializes the sequence generator and controller state.

• Clocking: clock/control logic supplies the active clock or gated clock to the pattern generator and signature register.

• Pattern application: LFSR outputs are mapped to two 4-bit adder operands. Carry-in is held at logic 0 in the documented configuration.

• Response compaction: S1, S2, S3, S4 and C4 are combined through an XOR parity network. The fault switch can invert S1 before compaction.

• Signature accumulation: the compacted parity bit is shifted into the signature register for each test cycle.

• Termination and comparison: the end-of-test detector stops further clocking after the documented 127-cycle run. The final signature is compared with 1110.

• Indication: a match produces PASS; a mismatch produces FAIL.

6. Components and functional roles

The bill of materials below is taken from the supplied project report and presentation. IC identifiers U1–U11 follow the reference documents.

ID	Part number	Functional role	Project use
U1	NE555	Clock generator	Adjustable master clock for discrete demonstrator
U2	74HC164	LFSR / pattern shift register	Generates test sequence and operand bits
U3	74LS83A	4-bit binary full adder	Circuit under test
U4, U5	74LS86 (2 ICs)	Quad XOR gates	Fault injection and response parity compaction
U6	74HC164	Signature register	Stores compacted response sequence
U7	74LS76A	Dual JK flip-flop	RUN and DONE control states
U8	74LS27	Triple 3-input NOR	End-of-test decode / control logic
U9	74LS11	Triple 3-input AND	End-of-test decode / finish logic
U10	74LS00	Quad 2-input NAND	Clock gating and result logic
U11	74LS85	4-bit magnitude comparator	Compares signature against golden code

6.1 Functional description of major blocks

NE555 clock: In the discrete circuit, the NE555 is configured as an oscillator to generate the master clock. The supplied report gives an adjustable frequency range of approximately 7 Hz to 140 Hz. The clock rate affects test duration but not the logical sequence.

74HC164 pattern register: The 8-bit serial-in, parallel-out shift register provides the pattern-generator stages. Selected outputs are mapped to adder inputs A1–A4 and B1–B4. The supplied material describes a 127-pattern sequence. The exact feedback taps and seed must match the final circuit implementation.

74LS83A adder: This is the CUT. It accepts two 4-bit operands and a carry-in and produces four sum outputs plus carry-out. In the documented configuration, carry-in C0 is connected to 0 V.

74LS86 XOR network: XOR gates form the response parity tree. One XOR path is also used to demonstrate a fault by conditionally inverting S1 when FLT is active.

74HC164 signature register: The compacted parity stream is shifted into the register. Four stages provide signature bits SIG0 through SIG3.

Control logic: 74LS76A, 74LS27, 74LS11 and 74LS00 devices implement run/done state storage, terminal-state decoding, clock gating and output qualification in the discrete implementation.

74LS85 comparator: The comparator checks the accumulated signature against golden-code switches G0–G3. Equality indicates a match, subject to the limitations of signature-based testing.

7. IC pin configurations

The following pin information is transcribed from the supplied 12-page pinout reference. It is a readable pinout appendix for the documented discrete-IC design. Verify the manufacturer datasheet and package orientation before wiring. Pinout labels are not a substitute for a complete netlist.

U1 — NE555 clock generator

Pin	IC signal	Connection / net label
1	GND	0 V
2	TRIG	T/Th
3	OUT	CLK
4	RESET	+5 V
5	CTRL	CTRL
6	THRES	T/Th
7	DISCH	DISCH
8	VCC	+5 V

U2 — 74HC164 LFSR pattern generator

Pin	IC signal	Connection / net label
1	A	FB
2	B	+5 V
3	QA	Q1
4	QB	Q2
5	QC	Q3
6	QD	Q4
7	GND	0 V
8	CP	CLKG
9	/MR	/START
10	QE	Q5
11	QF	Q6
12	QG	Q7
13	QH	Q8
14	VCC	+5 V

U3 — 74LS83A adder under test

Pin	IC signal	Connection / net label
1	A4	Q4
2	S3	S3
3	A3	Q3
4	B3	Q7
5	VCC	+5 V
6	S2	S2
7	B2	Q6
8	A2	Q2
9	S1	S1
10	A1	Q1
11	B1	Q5
12	GND	0 V
13	C0	0 V
14	C4	C4
15	S4	S4
16	B4	Q8

U4 — 74LS86 XOR, stage 1

Pin	IC signal	Connection / net label
1	1A	Q1
2	1B	nQ7
3	1Y	FB
4	2A	S1
5	2B	FLT
6	2Y	S1F
7	GND	0 V
8	3Y	X1
9	3A	S1F
10	3B	S2
11	4Y	X2
12	4A	S3
13	4B	S4
14	VCC	+5 V

U5 — 74LS86 XOR, stage 2

Pin	IC signal	Connection / net label
1	1A	X1
2	1B	X2
3	1Y	X3
4	2A	X3
5	2B	C4
6	2Y	P
7	GND	0 V
8	3Y	F03
9	3A	SIG0
10	3B	SIG3
11	4Y	DIN
12	4A	P
13	4B	F03
14	VCC	+5 V

U6 — 74HC164 signature register

Pin	IC signal	Connection / net label
1	A	DIN
2	B	+5 V
3	QA	SIG0
4	QB	SIG1
5	QC	SIG2
6	QD	SIG3
7	GND	0 V
8	CP	CLKG
9	/MR	/START
10	QE	NC
11	QF	NC
12	QG	NC
13	QH	NC
14	VCC	+5 V

U7 — 74LS76A RUN / DONE flip-flops

Pin	IC signal	Connection / net label
1	1CLK	CLK
2	1/PRE	/START
3	1/CLR	+5 V
4	1J	0 V
5	VCC	+5 V
6	2CLK	CLK
7	2/PRE	+5 V
8	2/CLR	/START
9	2J	END
10	2Q'	NC
11	2Q	DONE
12	2K	0 V
13	GND	0 V
14	1Q'	nRUN
15	1Q	RUN
16	1K	DONE

U8 — 74LS27 NOR gates

Pin	IC signal	Connection / net label
1	1A	Q1
2	1B	Q2
3	2A	Q4
4	2B	Q5
5	2C	0 V
6	2Y	N2
7	GND	0 V
8	3Y	nQ7
9	3A	Q7
10	3B	Q7
11	3C	Q7
12	1Y	N1
13	1C	Q3
14	VCC	+5 V

U9 — 74LS11 AND gates

Pin	IC signal	Connection / net label
1	1A	N1
2	1B	N2
3	2A	E1
4	2B	Q7
5	2C	+5 V
6	2Y	END
7	GND	0 V
8	3Y	FINISH
9	3A	DONE
10	3B	nRUN
11	3C	+5 V
12	1Y	E1
13	1C	Q6
14	VCC	+5 V

U10 — 74LS00 NAND gates

Pin	IC signal	Connection / net label
1	1A	CLK
2	1B	RUN
3	1Y	CLKG
4	2A	PASS_raw
5	2B	PASS_raw
6	2Y	nPASS
7	GND	0 V
8	3Y	nGREEN
9	3A	FINISH
10	3B	PASS_raw
11	4Y	nRED
12	4A	FINISH
13	4B	nPASS
14	VCC	+5 V

U11 — 74LS85 comparator

Pin	IC signal	Connection / net label
1	B3	G3
2	I(A less than B)	0 V
3	I(A equals B)	+5 V
4	I(A greater than B)	0 V
5	O(A greater than B)	NC
6	O(A equals B)	PASS_raw
7	O(A less than B)	NC
8	GND	0 V
9	B0	G0
10	A0	SIG0
11	B1	G1
12	A1	SIG1
13	A2	SIG2
14	B2	G2
15	A3	SIG3
16	VCC	+5 V

8. Design equations and logic

8.1 Adder function

For each bit position i, the full-adder sum is the XOR of the two operand bits and the carry-in. The carry-out is generated by the majority function of the three inputs. The operation is repeated across four bit positions with carry propagation from the least significant stage to the most significant stage.

S■ = A■ ⊕ B■ ⊕ C■■■

C■ = A■B■ + A■C■■■ + B■C■■■

The documented test configuration ties C0 to logic 0. The CUT therefore evaluates the two 4-bit operands without an externally supplied initial carry.

8.2 Fault injection and parity compaction

The fault switch is described as a control that inverts S1 before it enters the parity tree. In the normal state, the response path is unchanged. When fault injection is enabled, the S1 contribution is inverted, changing the compacted response stream and therefore the final signature.

P = S1F ⊕ S2 ⊕ S3 ⊕ S4 ⊕ C4

Here S1F is the possibly fault-inverted S1 signal. The parity bit P is presented to the signature register as DIN. The supplied presentation shows intermediate XOR terms X1, X2, X3 and P; exact gate-level grouping follows the IC pinout reference.

8.3 Signature register

The 4-bit signature register receives the compacted bit once per active test clock. The supplied presentation describes a feedback connection involving the signature stages and a feedback polynomial. The final golden signature for the documented fault-free sequence is 1110, with bit order SIG3 SIG2 SIG1 SIG0. The precise RTL shift direction and feedback expression should be taken from the final Verilog source when reproducing the implementation.

8.4 Golden signature comparison

The 74LS85 comparator compares the accumulated signature with the four golden-code switch inputs. In the documented setup, G3 G2 G1 G0 are configured to 1110. Equality indicates that the observed signature matches the expected reference. A mismatch is treated as a fault indication. Result logic qualifies the indication with the test-finished state so the LEDs represent a completed test.

9. Test methodology and fault model

The reference report describes three verification stages: a fault-free baseline run; a manual FLT-switch test that inverts S1; and structural stuck-at conditions on adder inputs and outputs. The Verilog testbench additionally checks a deliberately incorrect golden reference.

Test stage	Setup	Expected observation
Baseline fault-free	FLT = GND; golden = 1110	Signature 1110; PASS
Live fault injection	FLT = +5 V; S1 inverted	Signature differs from 1110; FAIL
Wrong golden (RTL test)	Golden reference set to 1111	Actual signature 1110; comparison reports FAIL
Structural stuck-at study	Selected adder inputs / outputs forced to 0 or 1	Signature differs from golden in reported cases

9.1 Stuck-at fault model

A stuck-at-0 (SA0) fault models a signal that remains at logic 0 regardless of the value it should carry. A stuck-at-1 (SA1) fault models a signal that remains at logic 1. The supplied report states that 31 single stuck-at conditions were evaluated across adder inputs A1–A4 and B1–B4 and outputs S1–S4 and C4. It reports that all 31 evaluated conditions produced signatures different from the golden code.

9.2 Fault coverage and aliasing

The source report gives a tested fault-detection result of 31 out of 31, or 100%, for the specific single-fault cases evaluated. This is not a claim that every possible fault in every implementation will always be detected. The 4-bit signature has a theoretical aliasing estimate of 1/16 = 6.25% under the simplified random-error model. The 127-pattern sequence does not exhaustively enumerate all 256 possible combinations of the eight adder operand bits.

10. FPGA / Verilog implementation

The project was also organized as a synthesizable FPGA design in Vivado 2024.2. The supplied source set contained two design files, a testbench, an XDC constraints file and a Tcl project-creation script.

File	Purpose
bist_core.v	Core BIST logic: pattern generation, adder response processing, signature and decision logic.
bist_fpga_top.v	Top-level FPGA wrapper connecting the BIST core to board-level inputs and outputs.
tb_bist_fpga.v	Simulation testbench driving normal, injected-fault and incorrect-reference cases.
boolean_bist.xdc	FPGA package-pin and timing constraints. Exact package pin values must be taken from this file.
create_project-2.tcl	Tcl automation script used to create the Vivado project and add design, testbench and constraints.

10.1 Target device and project setup

The Vivado project was created by sourcing the Tcl script from the folder containing the five project files. The project selected Xilinx Spartan-7 device XC7S50CSGA324-1, as visible in the elaborated-design window. The top-level design was set to bist_fpga_top, while tb_bist_fpga was selected as the simulation top. The XDC file was added to the constraints fileset.

10.2 Simulation strategy

The testbench applies four scenarios: (1) a good adder with the expected golden signature, (2) fault injection enabled, (3) a deliberately incorrect golden code, and (4) a final good-adder run. The testbench reports signature, PASS and FAIL outputs, RGB indication and number of test ticks. The observed run completed at 42,105 ns and printed “ALL TESTS PASSED”.

10.3 Synthesis and implementation

The Vivado screenshots show synthesis completing successfully and implementation reaching the “Implementation Complete” state. The elaborated / implemented design view displayed 81 cells, 28 I/O ports and 213 nets. These are tool-reported summary counts from the captured project state, not a complete utilization report. A timing-summary report and exact XDC package-pin assignments should be retained with the final submission.

10.4 FPGA status boundary

The evidence confirms behavioral simulation, synthesis and implementation. It does not confirm bitstream generation, programming through Hardware Manager or physical testing on the Spartan-7 board. Those steps should be completed and documented before describing the FPGA prototype as physically validated.

11. Vivado simulation and implementation results

Test	Condition	Signature	PASS	FAIL	Ticks	Interpretation
1	Good adder; golden 1110	1110	1	0	127	Correct fault-free result
2	Fault injection ON	1101	0	1	127	Injected fault detected
3	Golden reference = 1111	1110	0	1	127	Incorrect reference causes mismatch
4	Good adder again	1110	1	0	127	Normal result restored

The simulation console also showed RGB (B G R) = 010 for the two PASS cases and 001 for the two FAIL cases. Each test ran for 127 ticks. The testbench’s “ALL TESTS PASSED” message means that observed outputs matched the testbench’s expected outcomes; it does not mean every test case should display PASS on the DUT. In the injected-fault and wrong-golden cases, the expected DUT behavior is FAIL.

11.1 Synthesis / implementation observations

Synthesis completed successfully in Vivado 2024.2. The subsequent implementation run also completed successfully. The schematic view displayed 81 cells, 28 I/O ports and 213 nets. The captured screenshot does not show numerical setup / hold slack values. Timing closure should therefore be confirmed from the Timing Summary report, with WNS and WHS non-negative and all relevant paths constrained.

12. Results analysis and limitations

The testbench results support the intended sequence of operation: a fault-free adder produces signature 1110; the injected fault changes the signature to 1101 and asserts FAIL; a deliberately wrong golden code causes a mismatch even though the observed signature remains 1110; and the final normal run returns to PASS. This verifies the response path and comparator sensitivity to the selected golden reference.

The supplied report also records these signatures for stuck-at tests:

Faulted output	SA0 signature	SA1 signature	Reported outcome
S1	1001	1010	FAULT DETECTED
S2	1000	1011	FAULT DETECTED
S3	0000	0011	FAULT DETECTED
S4	0100	0111	FAULT DETECTED
C4	0101	0110	FAULT DETECTED

12.1 Limitations

• A 4-bit signature is compact but has a non-zero probability of aliasing; 6.25% is a theoretical estimate under a simplified random-error assumption.

• The documented 127-pattern sequence covers only part of the 256 possible 8-bit operand combinations.

• The supplied documents describe LFSR and signature feedback functionally; exact RTL feedback equations and seed should be checked against final source files.

• The discrete IC pinout reference is not the FPGA package pinout. FPGA pin assignments must be taken from boolean_bist.xdc and checked against the exact board revision.

• The available Vivado screenshots confirm simulation, synthesis and implementation, but not bitstream programming or physical FPGA operation.

• Reported 31/31 stuck-at coverage applies only to the 31 fault cases described in the reference report.

13. Applications and future enhancements

13.1 Applications

• Educational demonstration of design-for-testability and BIST principles.

• Testing of arithmetic datapaths and small combinational logic blocks.

• Embedded self-checking functions where external test access is limited.

• Demonstration of pseudo-random pattern generation, response compaction and signature comparison.

• FPGA-based prototyping of test controllers before integration into larger digital designs.

13.2 Future enhancements

Wider signature: Expand the signature analyzer from 4 bits to 8 bits. Under the simplified random-alias model, an 8-bit signature has an estimated aliasing probability of 1/256 = 0.390625%.

MISR architecture: Replace the simple serial signature register with a Multiple-Input Signature Register (MISR) for scalable response compaction.

Improved test coverage: Increase the number and diversity of test patterns, including deterministic vectors targeted at specific fault classes.

On-board FPGA verification: Generate the bitstream, program the target Spartan-7 board, verify switch / button / LED mappings and record measured hardware behavior.

Timing and resource reporting: Include utilization, timing summary, WNS / WHS, clock constraints and implementation warnings in the final submission.

Automated regression: Extend the testbench to cover reset behavior, repeated start requests, terminal count, fault-switch transitions and invalid golden signatures.

14. Conclusion

A BIST architecture for a 4-bit binary adder has been described using an LFSR-based pattern generator, a 74LS83A circuit under test, an XOR response compactor, a 4-bit signature register and a golden-signature comparator. The documented discrete design demonstrates autonomous test sequencing, response compaction, end-of-test control and visible PASS / FAIL reporting. The reference report records a golden signature of 1110, a 127-cycle test sequence and detection of all 31 single stuck-at cases evaluated.

The FPGA-oriented implementation was processed in Vivado 2024.2 for the Spartan-7 XC7S50CSGA324-1 target. The captured behavioral simulation completed all four testbench scenarios with the expected signatures and status outputs. Synthesis and implementation were also reported complete. These results establish successful RTL simulation and FPGA implementation-flow progress. Physical FPGA validation remains a milestone until bitstream generation, device programming and board-level testing are documented.

Appendix A. Practical implementation checklist

Step	Action	Evidence to retain
1	Confirm bist_fpga_top is design top and tb_bist_fpga is simulation top.	Vivado Sources screenshot
2	Run behavioral simulation and confirm all four testbench scenarios.	Console transcript and waveform
3	Run synthesis.	Synthesis completion and utilization report
4	Run implementation.	Implementation completion status
5	Open Timing Summary and check WNS, WHS and unconstrained paths.	Timing Summary report
6	Verify XDC pin locations against exact board manual / revision.	XDC file and board pinout reference
7	Generate bitstream.	Bitstream generation completion
8	Program FPGA using Hardware Manager.	Programming screenshot
9	Test START, fault input, golden switches and PASS / FAIL indicators.	Photographs and measured observations

Appendix B. Notes on signal names and pinout use

The IC pinout tables preserve the net labels shown in the supplied pinout document. Some labels are abbreviated in the source diagrams (DIN, CLKG, SIG0–SIG3, FINISH, PASS_raw and nRUN). Signal polarity is indicated by a leading slash or “n” where shown. These circuit-specific connection maps should be checked against the final schematic and physical wiring before construction.

The FPGA XDC file uses package-pin identifiers for the selected board. Those package pins are separate from the DIP pin numbers in the 74-series IC tables. Do not transfer IC pin numbers into the FPGA constraints file.

Appendix C. References

[1] BIST 4-Bit Adder Project Report, supplied reference document, 3 pages. Includes aim, block architecture, test methodology, experimental results, fault-coverage analysis and conclusion.

[2] BIST 4-Bit Adder Presentation, supplied reference document, 12 slides. Includes system overview, architecture, IC bill of materials, LFSR, fault injection, signature analysis, control logic, comparator, fault coverage and future scope.

[3] IC Pinout and Connection Reference, supplied document, 12 pages. Includes overall functional flow and pin-level connection labels for U1–U11.

[4] Project implementation files: bist_core.v, bist_fpga_top.v, tb_bist_fpga.v, boolean_bist.xdc and create_project-2.tcl, as used in the Vivado 2024.2 workflow.

[5] Vivado 2024.2 screenshots supplied during the project workflow: behavioral simulation output, synthesis completion and implementation completion for the Spartan-7 XC7S50CSGA324-1 target.

END OF REPORT

Addendum: Vivado Simulation and Bitstream Evidence

This addendum incorporates the two additional Vivado 2024.2 screenshots supplied after preparation of the main report. These images update the implementation evidence and supersede the earlier provisional status notes where stated below.

A. Behavioral simulation waveform

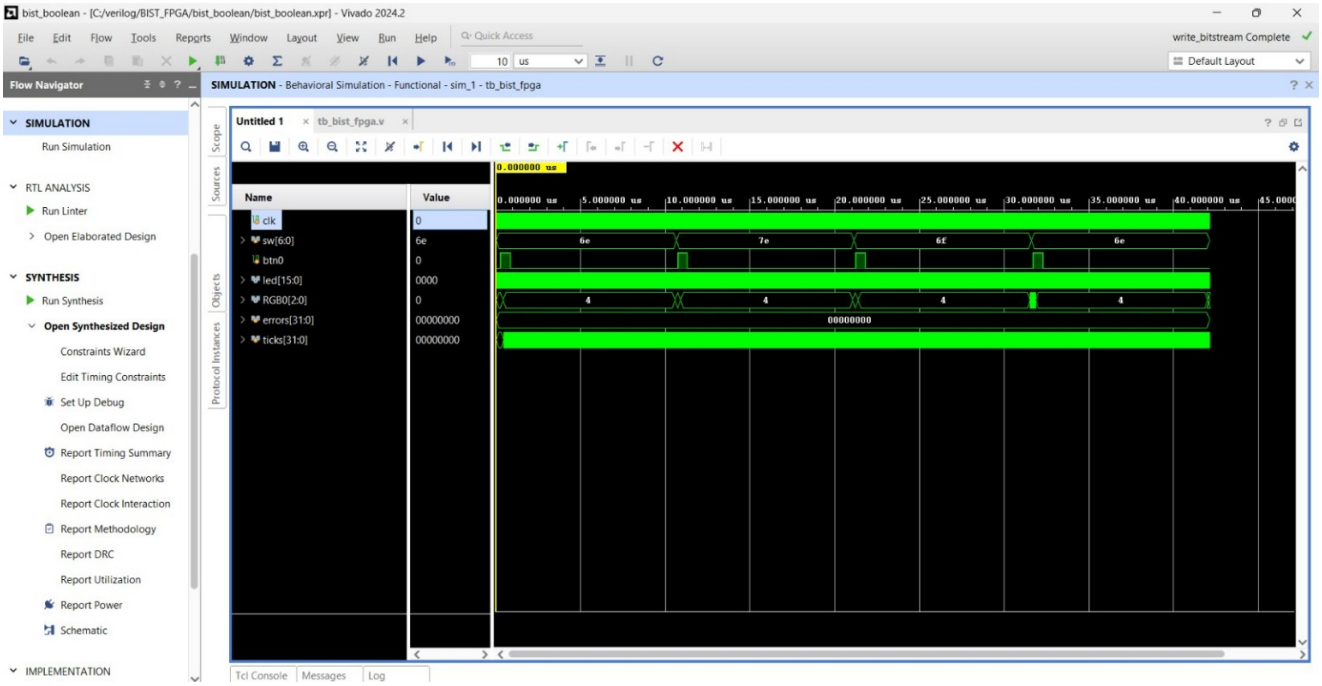


Figure A1. Vivado Behavioral Simulation — waveform view for tb_bist_fpga. The displayed signals include clk, sw[6:0], btn0, led[15:0], RGB[2:0], errors[31:0] and ticks[31:0]. The waveform shows the switch bus changing during the displayed interval; the value panel at the selected time shows clk = 0, sw = 6e, btn0 = 0, led = 0000, RGB = 4, errors = 0 and ticks = 0. These are cursor-time values, not a summary of the entire simulation.

Interpretation: The screenshot documents the simulation waveform and the internal / top-level signals being observed. The image alone does not establish a complete PASS/FAIL test sequence; the testbench console results documented in the main report remain the source for the four scenario outcomes.

B. Synthesized design schematic and updated tool status

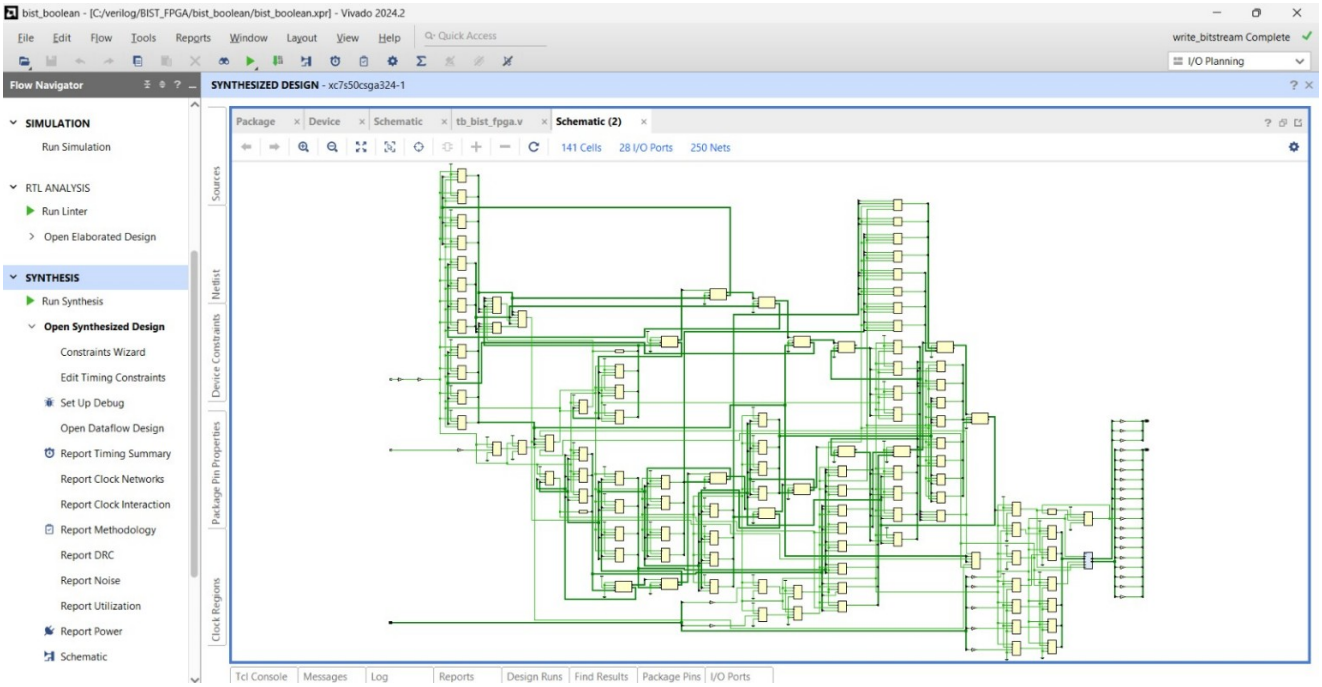


Figure A2. Vivado Synthesized Design schematic for the Spartan-7 target. The Vivado view reports 141 Cells, 28 I/O Ports and 250 Nets. The application title/status area also shows "write_bitstream Complete".

Updated implementation evidence: The latest screenshot confirms successful bitstream-generation completion in Vivado. It also updates the synthesized-design summary to 141 cells, 28 I/O ports and 250 nets. These figures supersede the earlier counts (81 cells and 213 nets) stated in Section 10.3 of the original report.

Physical validation boundary: "write_bitstream Complete" confirms that Vivado generated the programming bitstream. It does not, by itself, prove that the FPGA board was programmed or that the circuit was tested on the physical Spartan-7 hardware. Board programming and hardware verification should be documented separately.

Correction to the original report

The earlier implementation-status note and Section 10.4 stated that bitstream generation was not confirmed based on the evidence then available. The newly supplied screenshot supersedes that statement: bitstream generation is now confirmed. Physical device programming and board-level validation remain unconfirmed by the supplied screenshots.